LABORATORY · Lab · 26 hours

Experiment 1: Step Response of a Second-Order System



Control Systems Laboratory (22EEG53)

Control Systems (22EEG53)
Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
NMAM Institute of Technology, Nitte · 2025-26

LAB

Learning objectives

- Obtain the step response for ζ = 0, underdamped, critically damped and overdamped cases
- Observe the effect of ζ on overshoot and oscillation
- Verify the response with the theoretical expressions

Syllabus context

Laboratory component of 22EEG53 Control Systems (0:0:2 hours per week). CIE: record and observation 25, viva-voce 5, lab test 20 marks.

LAB

Aim and requirements

Aim: to obtain the step response of a typical second-order system for undamped, underdamped, overdamped and critically damped cases using MATLAB/Simulink.

Software: MATLAB with the Control System Toolbox (Simulink where required).

Hardware: Personal computer.

System: $G(s) = \omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$, $\omega_n = 5$ rad/s.

LAB

Theory: damping cases

ζ	Poles	Response
0	±jωn	Undamped: sustained oscillation
0 < ζ < 1	−ζωn ± jωd	Underdamped: overshoot, decaying oscillation
1	−ωn (double)	Critically damped: fastest, no overshoot
> 1	Real, distinct	Overdamped: slow, no overshoot

LAB

Procedure

1

Define

Enter ωn and the set of ζ values.

2

Create TF

Use tf(wn^2, [1 2*zeta*wn wn^2]).

3

Simulate

Use step(sys, t) on a common time axis.

4

Record

Read Mp, tp, ts for the underdamped case; save plots.

Simulink alternative: Step → Transfer Fcn → Scope.

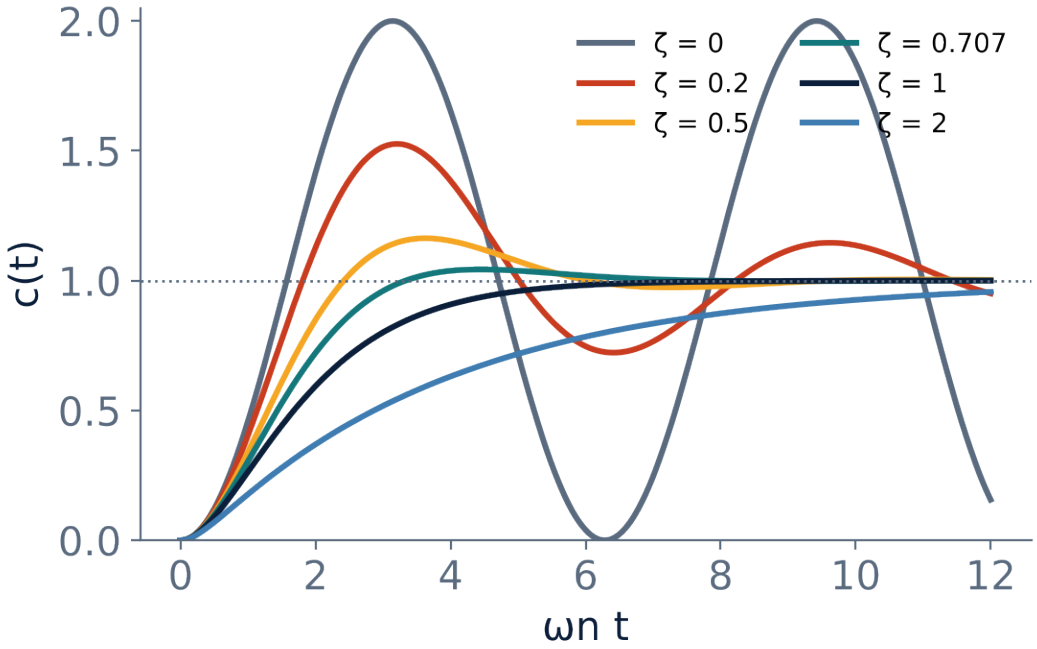
LAB

MATLAB program

```
wn = 5;  zeta = [0 0.3 0.7 1 2];
t = 0:0.01:6;  hold on
for k = 1:length(zeta)
  sys = tf(wn^2, [1 2*zeta(k)*wn wn^2]);
  y = step(sys, t);  plot(t, y, 'LineWidth', 1.5)
end
grid on; xlabel('Time (s)'); ylabel('c(t)')
legend('\zeta=0','0.3','0.7','1','2')
title('Step response of a second-order system')
```

LAB

Expected results



- ζ = 0: oscillates between 0 and 2.
- ζ = 0.3: Mp ≈ 37 %.
- ζ = 0.7: Mp ≈ 4.6 %.
- ζ = 1: no overshoot, fastest.
- ζ = 2: slow.

LAB

Observation table

ζ	Type	Mp (%) measured	Mp (%) theory	ts (s)
0	Undamped	100	100	∞
0.3	Under		37.2	
0.7	Under		4.6	
1	Critical	0	0	
2	Over	0	0	

LAB

Key takeaways

- ζ controls overshoot: small ζ → large overshoot
- ζ = 1 is the fastest response without overshoot
- MATLAB step() gives responses for all four cases
- Measured values agree with theory

LAB

Practice questions

Q1. What happens to the poles as ζ increases from 0 to 2?

Q2. Why does the ζ = 0 response not settle?

Q3. Write the MATLAB commands to get the step response.

Q4. For ωn = 5 and ζ = 0.5, find tp and Mp.